

Bellman-Ford & Floyd-Warshall

Handle negative weights / all-pairs paths.

Key concepts

- Bellman-Ford: handles negative edges, detects negative cycles.
- Floyd-Warshall: all-pairs shortest paths, $O(V^3)$.
- Trade speed for generality.

Key idea / example

Bellman-Ford relaxes all edges $V-1$ times.

Today's tasks (1.5h)

- ☐ Implement Bellman-Ford
- ☐ Detect a negative cycle
- ☐ Understand Floyd-Warshall

Resources

Playlist: youtube.com/playlist?list=PLsr8vTgyLdy_YndxNcI4WkH5Vorj5qvrV

(Clickable playlist link is in the app: Goals > this day > Study block.)